The kink without centring: Theorem 1 for a pre-norm RMSNorm block

Note for the LayerNorm-kink study (Q7a, 4 November 2026)

1 Setting

The surrogate's first layer is $h = Ez + b$ with $z = (x, u) \in \mathbb{R}^2$, $E \in \mathbb{R}^{H\times 2}$ of rank 2, $b \in \mathbb{R}^H$, $H = 16$. LayerNorm normalises the centred vector Ph, $P = I - \mathbf{1}\mathbf{1}^\top/H$. RMSNorm omits the centring:

$$\mathrm{RMS}(h) = \frac{h}{\sqrt{\|h\|^2/H + \varepsilon}}, \qquad v = \gamma \odot \mathrm{RMS}(h) + \beta.$$

Theorem 1 was stated with $Q_E = PE$ and $c = Pb$. The proposition below is the same statement with P replaced by I.

2 Reduction

Let $z^* = -(E^\top E)^{-1}E^\top b$ (the point where $\|Ez+b\|$ is smallest), $c_\perp = Ez^* + b$ (the part of b orthogonal to the range of E), $D^2 = \|c_\perp\|^2 + H\varepsilon$, and $E = U\Sigma V^\top$ a thin singular value decomposition ($U \in \mathbb{R}^{H\times 2}$).

Proposition 1. *For every z, with $s = z - z^*$ and $w = \Sigma V^\top s = U^\top Es$,*

$$\mathrm{RMS}(Ez + b) = \sqrt{H}\,\frac{Uw + c_\perp}{\sqrt{\|w\|^2 + D^2}}.$$

Proof. $Ez + b = Es + c_\perp$. Since $c_\perp \perp \mathrm{range}(E)$, $\|Ez + b\|^2 = \|Es\|^2 + \|c_\perp\|^2$ and $\|Es\| = \|w\|$. Then $\|h\|^2/H + \varepsilon = (\|w\|^2 + D^2)/H$. □

So the RMSNorm feature map has exactly the structure of Theorem 1: a centre z^*, a kink size D, half-widths $r_x^* = D/\|e_x\|$ and $r_u^* = D/\|e_u\|$ (columns of E), peak feature gain $\sqrt{H}\,\sigma_{\max}(E)/D$ at z^*, and the same far-field decay. The only change is which part of the bias sets the kink size: all of b outside the range of E, including its mean.

Corollary 1 (comparison with LayerNorm for the same weights)**.** *For the same (E, b), $\|c_\perp^{\mathrm{RMS}}\| \geq \|c_\perp^{\mathrm{LN}}\|$, hence $D_{\mathrm{RMS}} \geq D_{\mathrm{LN}}$: without centring the kink is never narrower.*

Proof. $\|c_\perp^{\mathrm{LN}}\| = \min_z \|P(Ez+b)\| \leq \min_z \|Ez+b\| = \|c_\perp^{\mathrm{RMS}}\|$ because P is an orthogonal projection. □

Equality needs the residual $Ez^*_{\mathrm{RMS}} + b$ to have zero mean and the LN minimiser to coincide with it. The difference comes from the mean of the bias (and of the encoder columns). With the encoder bias initialised at 0, both $c_\perp$ start at 0 and grow with training.

3 Numerical check

`theory/verify_rms.py` (float64): over 300 random (E, b, ε) with $\varepsilon \in \{0, 10^{-6}, 10^{-2}\}$ the reduction holds to 8.9×10^{-16} (the Uw form to 1.8×10^{-15}); the largest singular value of $\partial\,\mathrm{RMS}/\partial z$ at z^* equals $\sqrt{H}\sigma_{\max}(E)/D$ to 4×10^{-9} (central differences); the corollary holds on all 300 draws.

Applied to the trained LN MSE 3000 encoders (read with the RMS formula, the same (E, b)), the two kink sizes differ by at most 4 %: D_{LN} from 0.132 to 0.901 and D_{RMS} from 0.137 to 0.908; the trained biases have mean $|\bar{b}| \leq 0.027$, so almost all of $\|b\|$ is already orthogonal to $\mathbf{1}$.

4 What the theory does and does not predict for trained RMSNorm surrogates

- It predicts the same kink structure (a centre, a size D, slope errors of order $1/D$) at the point where $\|Ez + b\|$ is smallest. Since the bias starts at 0, the centre starts at the origin of (x, u), as for LayerNorm.
- For identical weights, $D_{\mathrm{RMS}} \geq D_{\mathrm{LN}}$. On LN-trained weights the difference is small, so the theory gives no reason to expect a qualitatively wider kink. Training an RMSNorm surrogate changes the weights, and the theory says nothing about how.
- It does not predict the failure rates: these depend on the trained D and on the slope errors MSE training leaves, as for LayerNorm.